

Khalsa College of Engineering and Technology

Amritsar, Punjab



PROJECT REPORT

Project Name: - CUSTOMER SHOPPING BEHAVIOR ANALYSIS

Semester:- 6th

Branch:- BCA

Submitted To:

Submitted By:

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CERTIFICATE

CUSTOMER SHOPPING BEHAVIOR ANALYSIS

This is to certify that the project titled "CUSTOMER SHOPPING BEHAVIOR ANALYSIS " is a Bonafide work carried out by Sonu Kumar students of Bachelor of Computer Applications (BCA), in partial fulfilment of the requirements for the award of the degree of Bachelor of Computer Applications (BCA) at Khalsa College of Engineering & Technology, Amritsar.

This project is the result of the students' own work and has not been submitted for any other degree or diploma.

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ACKNOWLEDGEMENT

I would like to express my sincere gratitude to all those who supported and guided me throughout the development of my project, "CUSTOMER SHOPPING BEHAVIOR ANALYSIS." This project would not have been possible without their valuable contributions.

First and foremost, I extend my heartfelt thanks to my project guide, Ms. Sukhmeet Kaur, for her invaluable guidance, constant encouragement, and technical expertise. Her insightful suggestions, patience, and constructive feedback helped me overcome challenges and successfully complete this project.

I would also like to express my sincere thanks to Ms. Shivani Mahajan, to whom this project is submitted, for their support, motivation, and valuable suggestions during the development of this project.

I am also deeply grateful to the Head of the Department of Computer Applications (BCA) for providing the necessary resources, infrastructure, and academic support throughout this endeavor. I would also like to thank all the faculty members of the BCA Department for their valuable teachings and continuous motivation.

I extend my appreciation to Khalsa College of Engineering & Technology, Amritsar, for providing a conducive environment and access to the necessary facilities that enabled me to complete this project successfully.

I am also thankful to the library staff for providing access to relevant books and resources, and to the technical staff for their assistance in maintaining the system infrastructure.

My sincere thanks go to my friends and classmates for their constant support, valuable discussions, and technical assistance during the development phase.

Finally, I owe my deepest gratitude to my family for their unwavering support, patience, and encouragement throughout this journey.

I am grateful to everyone who directly or indirectly contributed to the successful completion of this project.

DECLARATION

I hereby declare that the project titled " CUSTOMER SHOPPING BEHAVIOR ANALYSIS" submitted by Sonu Kumar (2304055) in partial fulfilment of the requirements for the award of the degree of Bachelor of Computer Applications (BCA) to Khalsa College of Engineering & Technology, Amritsar, is an original work carried out by us under the guidance of our project guides.

We further declare that this project has not been submitted previously, either in full or in part, to any other university or institution for the award of any degree or diploma. All the information and data included in this report are authentic to the best of our knowledge.

We also declare that all sources of information used in this project have been properly acknowledged in the references section.

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ABSTRACT

This project focuses on analyzing customer shopping behavior using transactional data consisting of 3,900 purchase records across multiple product categories. The primary objective of this study is to understand customer purchasing patterns, identify key factors influencing buying decisions, and provide actionable insights to improve business performance.

The dataset includes various features such as customer demographics (age, gender, location, subscription status), purchase details (product category, amount, season), and behavioral attributes like discount usage, review ratings, purchase frequency, and shipping preferences. During the data preparation phase, missing values in the review rating column were handled using appropriate imputation techniques, and new features such as age groups and purchase frequency were created to enhance the analysis.

Exploratory Data Analysis (EDA) was conducted using Python to uncover initial trends and patterns. Further, structured data analysis was performed using SQL to answer key business questions, including revenue contribution by different customer segments, the impact of discounts on spending behavior, product performance based on ratings, and differences between subscriber and non-subscriber customers. The analysis also included customer segmentation into new, returning, and loyal customers to better understand long-term customer value.

To effectively communicate the findings, an interactive dashboard was developed using Power BI, enabling visualization of important metrics such as revenue trends, customer segments, and product performance. The insights derived from this analysis suggest that factors like discounts, subscription status, and customer loyalty significantly influence purchasing behavior.

Based on these findings, several business recommendations were proposed, including improving customer loyalty programs, optimizing discount strategies, promoting topperforming products, and implementing targeted marketing campaigns. Overall, this project demonstrates how data-driven analysis can help organizations better understand customer behavior, enhance decision-making, and achieve sustainable business growth.

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PROJECT OVERVIEW

Customer shopping behavior analysis is a crucial aspect of modern business strategy, especially in the retail and e-commerce sectors. Companies today collect vast amounts of transactional data, but the real value lies in transforming this data into meaningful insights. This project focuses on analyzing customer shopping behavior using a dataset consisting of approximately 3,900 purchase records across various product categories.

The main objective of this project is to understand how customers interact with products, what factors influence their purchasing decisions, and how businesses can use this information to improve sales and customer satisfaction. The analysis includes studying demographic patterns, product preferences, purchase frequency, discount usage, and subscription behavior.

In today's competitive market, companies like Amazon and Flipkart rely heavily on such data analysis to personalize customer experiences and optimize pricing strategies. By analyzing customer behavior, businesses can identify high-value customers, predict future trends, and design targeted marketing campaigns.

This project combines tools such as Python, SQL, and Power BI to perform end-to-end data analysis, starting from data cleaning and preprocessing to visualization and insight generation. The findings from this analysis can help organizations make data-driven decisions, improve operational efficiency, and enhance customer engagement.

BUSINESS PROBLEM STATEMENT

Retail businesses often face challenges in understanding customer behavior due to the complexity and volume of data generated from daily transactions. Despite having access to large datasets, many organizations struggle to extract meaningful insights that can guide strategic decision-making.

The primary problem addressed in this project is to analyze customer shopping behavior and identify patterns that can help improve business performance. Companies need to understand which products are most popular, how discounts influence purchasing decisions, and which customer segments contribute the most to revenue.

Another key challenge is identifying loyal customers and understanding their buying patterns. Businesses also need to determine whether subscription-based customers behave differently compared to non-subscribers. Additionally, analyzing the impact of shipping methods and seasonal trends is important for optimizing operations.

The central business question of this project is:
“How can customer shopping data be utilized to identify trends, improve customer engagement, and increase overall revenue?”

By addressing this problem, the project aims to provide actionable insights that can help businesses improve marketing strategies, enhance customer retention, and maximize profitability.

DATASET SUMMARY

The dataset used in this project consists of 3,900 rows and 18 columns, representing customer transactions across multiple product categories.

Key Features of the Dataset:

- **Customer Demographics:** Includes attributes such as age, gender, location, and subscription status.
- **Purchase Details:** Contains information about items purchased, product categories, purchase amounts, season, size, and color.
- **Shopping Behavior:** Includes discount usage, promo codes, previous purchases, purchase frequency, review ratings, and shipping type.

Data Quality Issues:

During the initial exploration, it was observed that the dataset contained missing values in the review rating column (approximately 37 entries). These missing values needed to be handled carefully to ensure accurate analysis.

Additionally, some columns were found to be redundant, such as promo code usage, which overlapped with discount information. These issues were addressed during the data preprocessing stage.

Overall, the dataset provides a comprehensive view of customer behavior, enabling detailed analysis of purchasing patterns and trends.

Exploratory Data Analysis using Python

Exploratory Data Analysis (EDA) is a critical step in understanding the dataset and identifying patterns, trends, and anomalies. In this project, Python was used extensively for data cleaning, transformation, and analysis.

Data Loading and Exploration:

The dataset was imported using the pandas library. Initial exploration was performed using functions such as `.info()` and `.describe()` to understand the structure and statistical properties of the data.

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	39
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	22
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	NaN

Discount Applied	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
3900	3900	3900.000000	3900	3900
2	2	NaN	6	7
No	No	NaN	PayPal	Every 3 Months
2223	2223	NaN	677	584
NaN	NaN	25.351538	NaN	NaN
NaN	NaN	14.447125	NaN	NaN
NaN	NaN	1.000000	NaN	NaN
NaN	NaN	13.000000	NaN	NaN
NaN	NaN	25.000000	NaN	NaN
NaN	NaN	38.000000	NaN	NaN
NaN	NaN	50.000000	NaN	NaN

Handling Missing Values:

Missing values in the review rating column were handled using median imputation, ensuring that the dataset remained consistent and reliable.

Feature Engineering:

New features were created to enhance the analysis:

- Age groups were created by categorizing customers into different age ranges.
- Purchase frequency was calculated to understand how often customers make purchases.
-

Data Transformation:

Column names were standardized to improve readability and maintain consistency.

Redundant columns were removed to avoid duplication and confusion.

Insights from EDA:

- Younger customers showed higher purchase frequency.
- Discount usage significantly influenced buying behavior.
- Certain product categories performed better in specific seasons.

DATA ANALYSIS USING SQL (BUSINESS TRANSACTIONS)

After preprocessing, the cleaned dataset was loaded into PostgreSQL for structured analysis using SQL queries.

Key Analyses Performed:

1. **Revenue by Gender:** Compared total revenue generated by male and female customers.

	gender text	revenue numeric
1	Female	75191
2	Male	157890

2. **High-Spending Customers:** Identified customers who used discounts but still spent above average purchase amount.

	customer_id bigint	purchase_amount bigint
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90
10	22	62
11	24	88
Total rows: 839		Query complete 00:00:01

3. **Top Products:** Determined the top-rated and most purchased products.

	item_purchased text	Average Product Rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

4. **Shipping Type Comparison** – Compared average purchase amounts between Standard and Express shipping.

	shipping_type text	round numeric
1	Standard	58.46
2	Express	60.48

5. **Subscribers vs. Non-Subscribers** – Compared average spend and total revenue across subscription status.

	subscription_status text	total_customers bigint	avg_spend numeric	total_revenue numeric
1	Yes	1053	59.49	62645.00
2	No	2847	59.87	170436.00

6. **Discount-Dependent Products** – Identified 5 products with the highest percentage of discounted purchases.

	item_purchased text	discount_rate numeric
1	Hat	50.00
2	Sneakers	49.66
3	Coat	49.07
4	Sweater	48.17
5	Pants	47.37

7. **Customer Segmentation** – Classified customers into New, Returning, and Loyal segments based on purchase history.

	customer_segment text	Number of Customers bigint
1	Loyal	3116
2	New	83
3	Returning	701

8. **Top 3 Products per Category** – Listed the most purchased products within each category.

	item_rank bigint	category text	item_purchased text	total_orders bigint
1	1	Accessories	Jewelry	171
2	2	Accessories	Sunglasses	161
3	3	Accessories	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161

9. **Repeat Buyers & Subscriptions** – Checked whether customers with >5 purchases are more likely to subscribe.

	subscription_status text	repeat_buyers bigint
1	No	2518
2	Yes	958

10. **Revenue by Age Group** – Calculated total revenue contribution of each age group.

	age_group text	total_revenue numeric
1	Young Adult	62143
2	Middle-aged	59197
3	Adult	55978
4	Senior	55763

DASHBOARD (POWER BI)

To effectively present the findings, an interactive dashboard was created using Power BI.

Dashboard Features:

- Revenue trends visualization
- Customer segmentation analysis
- Product performance tracking
- Discount and subscription impact

The dashboard allows users to filter data based on different parameters such as age group, product category, and location. This helps stakeholders quickly understand key insights and make informed decisions.



BUSINESS RECOMMENDATIONS

- **Boost Subscriptions** – Promote exclusive benefits for subscribers.
- **Customer Loyalty Programs** – Reward repeat buyers to move them into the “Loyal” segment.
- **Review Discount Policy** – Balance sales boosts with margin control.
- **Product Positioning** – Highlight top-rated and best-selling products in campaigns.
- **Targeted Marketing** – Focus efforts on high-revenue age groups and express-shipping users.

CONCLUSION

This project on customer shopping behavior analysis successfully demonstrates how data-driven techniques can be used to extract meaningful insights from transactional data. By analyzing a dataset of customer purchases, the study provides a comprehensive understanding of how different factors such as demographics, product preferences, discount usage, and subscription status influence customer buying behavior.

Throughout the project, various tools and technologies such as Python, SQL, and Power BI were utilized to perform data cleaning, transformation, analysis, and visualization. The use of Python helped in handling missing values, creating new features, and preparing the dataset for further analysis. SQL played a crucial role in performing structured queries to answer important business questions, while Power BI enabled the creation of interactive dashboards for effective data visualization.

The analysis revealed several important insights, including the significant impact of discounts on purchase decisions, the higher contribution of loyal and repeat customers to overall revenue, and the variation in buying patterns across different age groups and product categories. It was also observed that subscription-based customers tend to spend more, indicating the importance of customer retention strategies.

This project highlights the importance of understanding customer behavior in today's competitive business environment. By leveraging data analytics, organizations can make informed decisions, optimize marketing strategies, and improve customer satisfaction. The findings of this analysis can help businesses identify high-value customers, enhance product positioning, and develop targeted campaigns to increase sales and profitability.

In conclusion, this project not only provides valuable business insights but also demonstrates the practical application of data analytics tools in solving realworld problems. It emphasizes that effective use of data can significantly improve business performance and support long-term growth.

TOOLS & TECHNOLOGIES USED

This project was completed using a combination of modern data analytics tools and technologies, each serving a specific purpose in the data analysis lifecycle.

1. Python (Pandas & NumPy)

Python was used as the primary programming language for data preprocessing and exploratory data analysis. The Pandas library was used for data manipulation, cleaning, and transformation, while NumPy was used for numerical computations. Python helped in handling missing values, performing feature engineering, and preparing the dataset for further analysis.

2. SQL (PostgreSQL)

Structured Query Language (SQL) was used to perform in-depth data analysis. PostgreSQL database was used to store the cleaned dataset and execute queries to extract meaningful insights. SQL helped in analyzing customer segments, revenue distribution, product performance, and purchase patterns through structured queries.

3. Power BI

Power BI was used to create an interactive and visually appealing dashboard. It helped in presenting key insights such as revenue trends, customer segmentation, and product analysis in a user-friendly manner. The dashboard allows filtering and dynamic exploration of data, making it easier for stakeholders to understand the findings.

4. Microsoft Excel

Excel was used for initial data review and basic data handling tasks. It was helpful in quickly viewing the dataset, performing simple calculations, and verifying results during different stages of the analysis.

5. Jupyter Notebook / VS Code

Jupyter Notebook or Visual Studio Code was used as the development environment for writing and executing Python code. It provided an interactive interface for performing data analysis and documenting the workflow.

REFERENCES

The following sources were referred to during the completion of this project:

Data Source:

- Customer Shopping Behavior Dataset (Kaggle)

Documentation & Learning Resources:

- Python Pandas Official Documentation
- NumPy Official Documentation
- PostgreSQL Documentation
- Microsoft Power BI Documentation

Online Learning Platforms:

- YouTube tutorials for Python, SQL, and Power BI
- Online articles and blogs related to data analysis and customer behavior